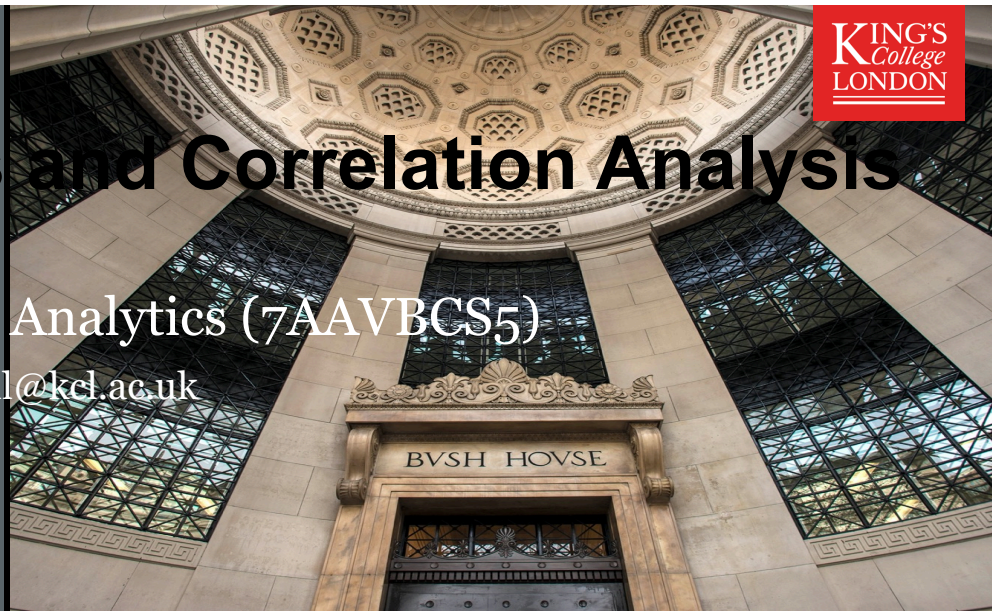


Scatter Plots and Correlation Analysis

Social and Cultural Analytics (7AAVBCS5)

Dr Mark J. Hill · mark.j.hill@kcl.ac.uk

2026-11-16



Lecture Plan

- **Part I: Introduction to Bivariate Analysis**
- **Part II: Scatter Plots**
- **Part III: Correlations**
- **Part IV: Creating Scatter Plots**
- **Part V: Thinking about causes** (*extra if time*)
- **Part VI: Live Coding**

Key Learning Outcomes

By the end of this week you will be able to:

- Explore the relationship between two numeric variables
- Read a scatter plot and describe the pattern
- Calculate and interpret Pearson and Spearman correlations
- Choose between them, and justify the choice

- Encode a third variable with size, colour or shape
- Explain why correlation is not causation

Review

- Data frames, selecting, filtering, missing data
- Categories, frequency tables, crosstabs, `groupby`
- Counts versus percentages
- Bar plots, distributions, histograms

This week: two numeric variables at once

Part I: Bivariate Analysis

Bivariate analysis

Bivariate analysis examines the relationship between two numeric/ordinal variables — how they might be connected or influence each other.

Moving beyond describing single variables to understanding how things **relate**.

Key questions

- Do the variables change together?
- When one increases, does the other increase or decrease?
- How strong is the relationship?
- Is there a pattern we can describe?

How we assess

1. **Scatter plots** — each point is one observation; position shows both values; the pattern reveals the relationship, or its absence
2. **Measures of correlation** — Pearson, Spearman,

regression models

Why relationships matter

In culture, phenomena rarely exist in isolation:

- Film budget ↔ box office revenue
- Social media engagement ↔ time of day
- Text length ↔ reading difficulty
- Historical period ↔ genre popularity

Why the double-headed arrow? **Collinearity** — relationships can work in either direction, or both. Statistics cannot tell you which. That judgement is yours as a researcher.

Part II: Scatter Plots

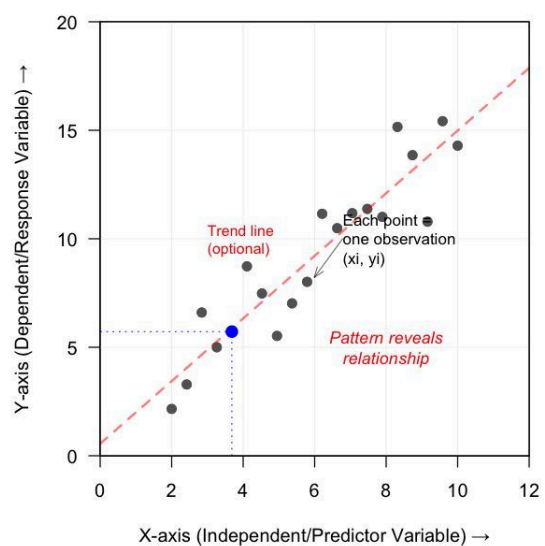
Anatomy of a scatter plot

- **X-axis:** independent / predictor variable
- **Y-axis:** dependent / response variable
- **Each point:** one observation
- **Pattern:** reveals the relationship

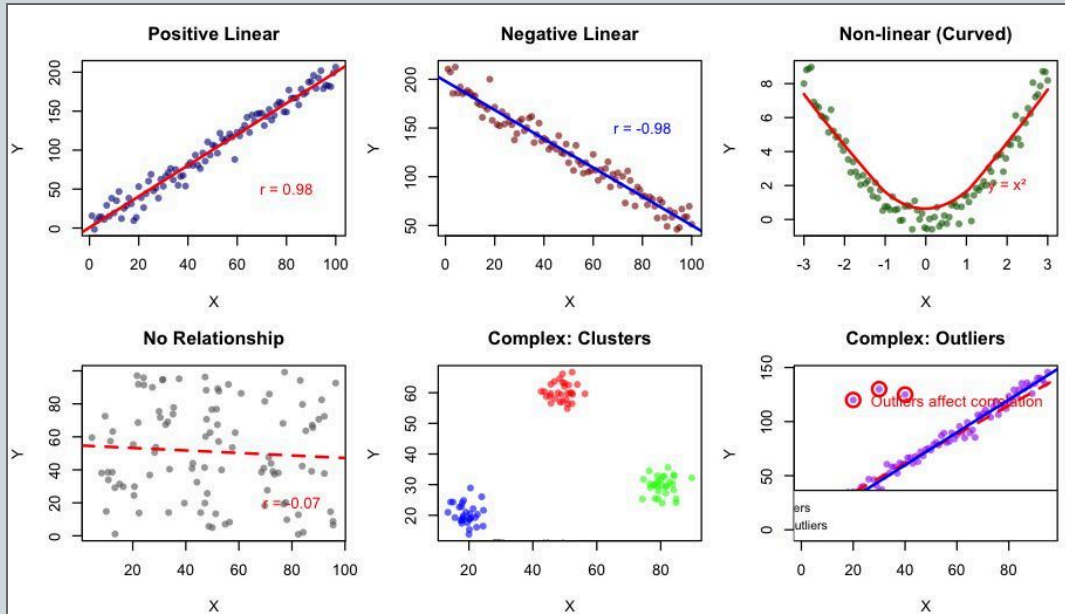
When to use

- To inspect a potential relationship between two numeric variables
- To verify statistical analyses — correlation, linear regression

Anatomy of a Scatter Plot



Types of relationships



Why visualise, when we can measure?

Because two datasets can have identical statistics and completely different shapes.

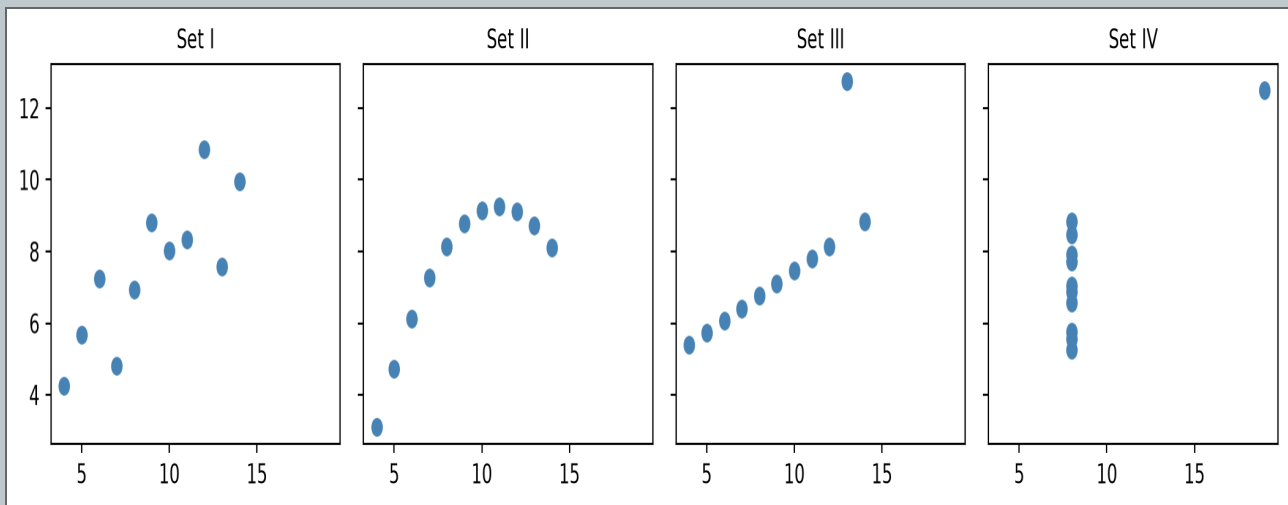
Anscombe's quartet

```
1 import pandas as pd
2
3 x1 = [10, 8, 13, 9, 11, 14, 6, 4, 12, 7, 5]
4 x4 = [8] * 8 + [19, 8, 8]
5 sets = {
6     "I": (x1, [8.04, 6.95, 7.58, 8.81, 8.33, 9.96, 7.24, 4.26, 10.84, 4.82, 5.68]),
7     "II": (x1, [9.14, 8.14, 8.74, 8.77, 9.26, 8.10, 6.13, 3.10, 9.13, 7.26, 4.74]),
8     "III": (x1, [7.46, 6.77, 12.74, 7.11, 7.81, 8.84, 6.08, 5.39, 8.15, 6.42, 5.73]),
9     "IV": (x4, [6.58, 5.76, 7.71, 8.84, 8.47, 7.04, 5.25, 5.56, 12.50, 7.91, 6.89]),
10 }
11
12 pd.DataFrame(
13     [{"set": k, "mean x": round(pd.Series(x).mean(), 2),
14      "mean y": round(pd.Series(y).mean(), 2),
15      "sd y": round(pd.Series(y).std(), 2),
16      "r": round(pd.Series(x).corr(pd.Series(y)), 3)}
17     for k, (x, y) in sets.items()]
18 )
```

	set	mean x	mean y	sd y	r
0	I	9.0	7.5	2.03	0.816
1	II	9.0	7.5	2.03	0.816
2	III	9.0	7.5	2.03	0.816
3	IV	9.0	7.5	2.03	0.817

The same four datasets, plotted

```
1 import matplotlib.pyplot as plt
2
3 fig, axes = plt.subplots(1, 4, figsize=(11, 2.7), sharex=True, sharey=True)
4 for ax, (name, (x, y)) in zip(axes, sets.items()):
5     ax.scatter(x, y, color="steelblue")
6     ax.set_title(f"Set {name}", fontsize=10)
7 plt.tight_layout()
8 plt.show()
```



Same mean. Same standard deviation. Same correlation. Four different stories.

Lesson: visualise as part of analysing

EDA is not a preliminary step you skip when busy.

Part III: Correlations

What is correlation?

Correlation measures two things:

1. **Strength** of association between two variables
 2. **Direction** of the relationship
- Coefficients range from **-1 to +1**:
- ± 1 = perfect association
 - ~ 0 = no relationship
 - > 0 = positive · < 0 = negative

Correlation $\neq$ Causation

OED, “coefficient”:

“Cooperating to produce a result.” “A joint agent or factor in producing an effect or result.”

Interpreting strength

r		Interpretation
0.8 to 1	−0.8 to −1	Very strong
0.6 to 0.8	−0.6 to −0.8	Strong
0.4 to 0.6	−0.4 to −0.6	Moderate
0.2 to 0.4	−0.2 to −0.4	Weak
0 to 0.2	0 to −0.2	Very weak / none

Note: the importance of an r value differs depending on the data. In some fields 0.3 is a major finding; in others 0.8 is unremarkable.

Pearson correlation

Measures the strength of a **straight-line** relationship.

- Uses your actual data values
- Best for “as X increases, Y increases in a straight line”

Strength: very precise when the relationship is linear

Weaknesses: one extreme value can throw it off badly; misses curves

When to use

- Both variables continuous
- The scatter plot looks roughly like a straight line
- No wild outliers

If you could draw a straight line through it, use Pearson.

Spearman correlation

Measures whether two variables **consistently move together**.

- Converts data to **rankings** first
- Captures any consistent pattern, even curves

Strengths: handles curves; outliers do not upset it; works with ordered categories such as Likert scales

Weakness: slightly less powerful than Pearson when the relationship really is linear

When to use

- Ordinal data (rankings, survey scales)
- Curved but consistent relationships
- Outliers present
- Unsure about the distribution

**Consistent movement, but
curvy or with outliers →
Spearman.**

What do we mean by rankings?

Original X	Ranked X	Original Y	Ranked Y
1	1	2	1
3	2	5	2
7	3	10	3
10	4	18	4
12	5	30	5
14	6	35	6
15	7	45	7

Spearman correlates the **ranks**, not the values. So it does not care how far apart the numbers are — only about their order.

Correlation in Python

```
1 dawtry = pd.read_csv("../data/DawtryEtAl2015.csv")
2
3 pair = dawtry[["Household_Income", "Social_Circle_Mean_Income"]].dropna()
4 print("n =", len(pair))
5 print("Pearson :", round(pair["Household_Income"].corr(pair["Social_Circle_Mean_Income"]
6 print("Spearman:", round(pair["Household_Income"].corr(pair["Social_Circle_Mean_Income"]
7                                     method="spearman"), 3))
```

```
n = 301
Pearson : 0.475
Spearman: 0.405
```

When Pearson and Spearman disagree

```
1 pair = dawtry[["Household_Income", "fairness"]].dropna()
2 print("Pearson :", round(pair["Household_Income"].corr(pair["fairness"]), 3))
3 print("Spearman:", round(pair["Household_Income"].corr(pair["fairness"], method="spearmanr"), 3))
```

```
Pearson : 0.174
Spearman: 0.234
```

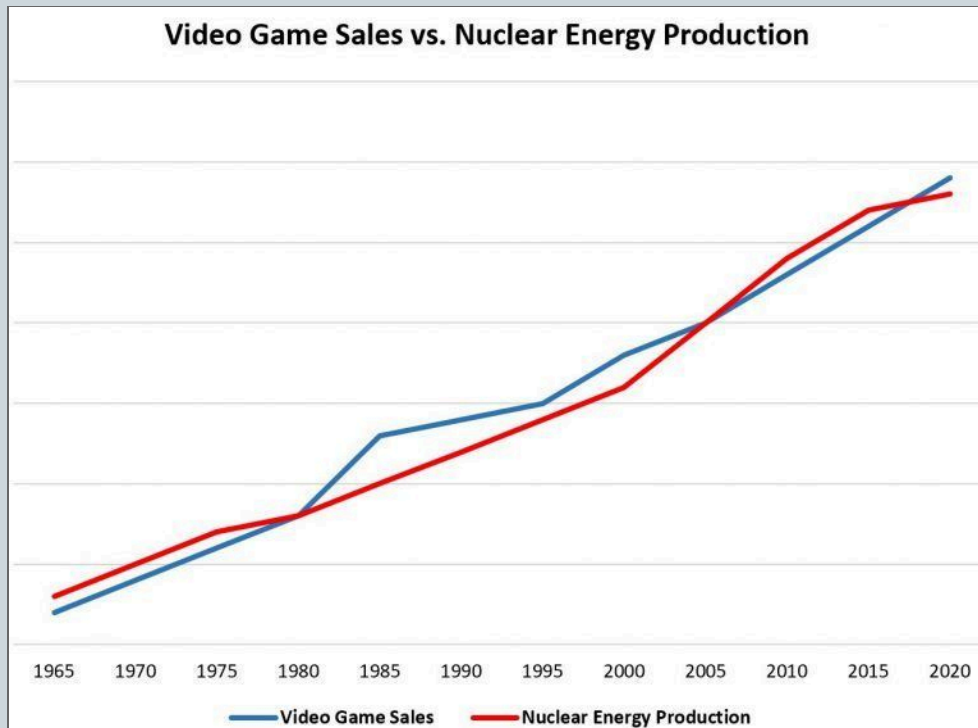
Pearson says 0.174 — very weak. Spearman says 0.234 — weak but noticeably stronger. Why?

A real correlation, and a real null

```
1 for a, b in [("Political_Preference", "redist1"), ("age", "Household_Income")]:
2     s = dawtry[[a, b]].dropna()
3     print(f"{a:22} x {b:18} n={len(s):3}   r = {s[a].corr(s[b]):+.3f}")
```

Political_Preference	x	redist1	n=301	r = -0.520
age	x	Household_Income	n=300	r = +0.018

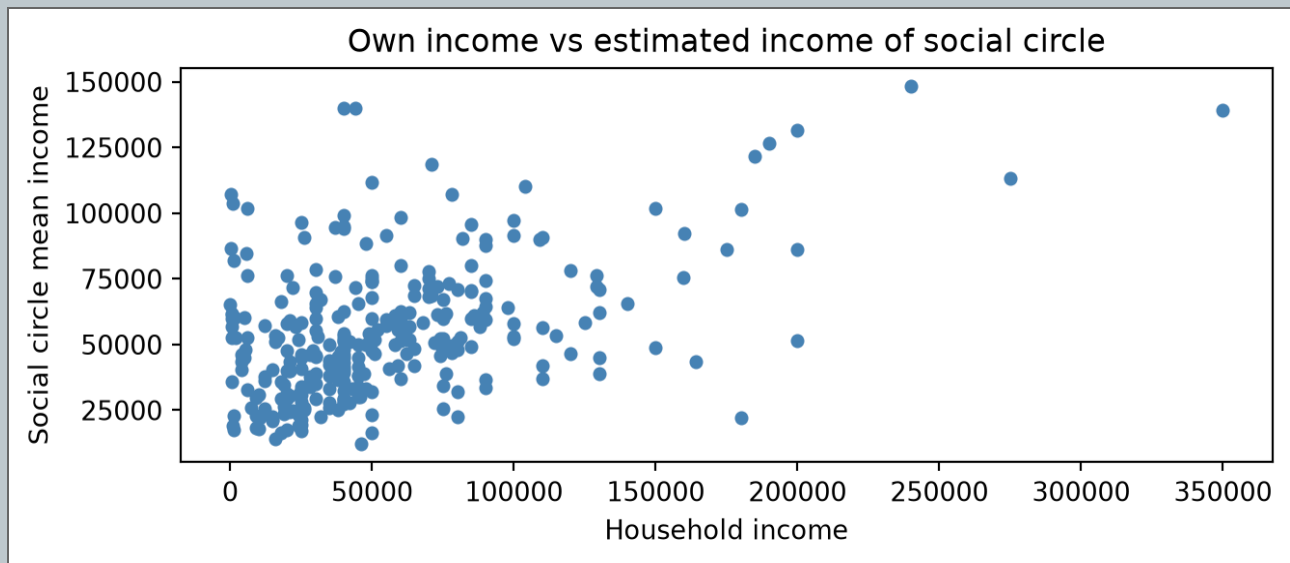
Bad correlations



Part IV: Creating scatter plots

A basic scatter plot

```
1 fig, ax = plt.subplots(figsize=(7, 3))
2 ax.scatter(dawtry["Household_Income"], dawtry["Social_Circle_Mean_Income"],
3           color="steelblue", s=18)
4 ax.set_title("Own income vs estimated income of social circle")
5 ax.set_xlabel("Household income")
6 ax.set_ylabel("Social circle mean income")
7 plt.tight_layout()
8 plt.show()
```



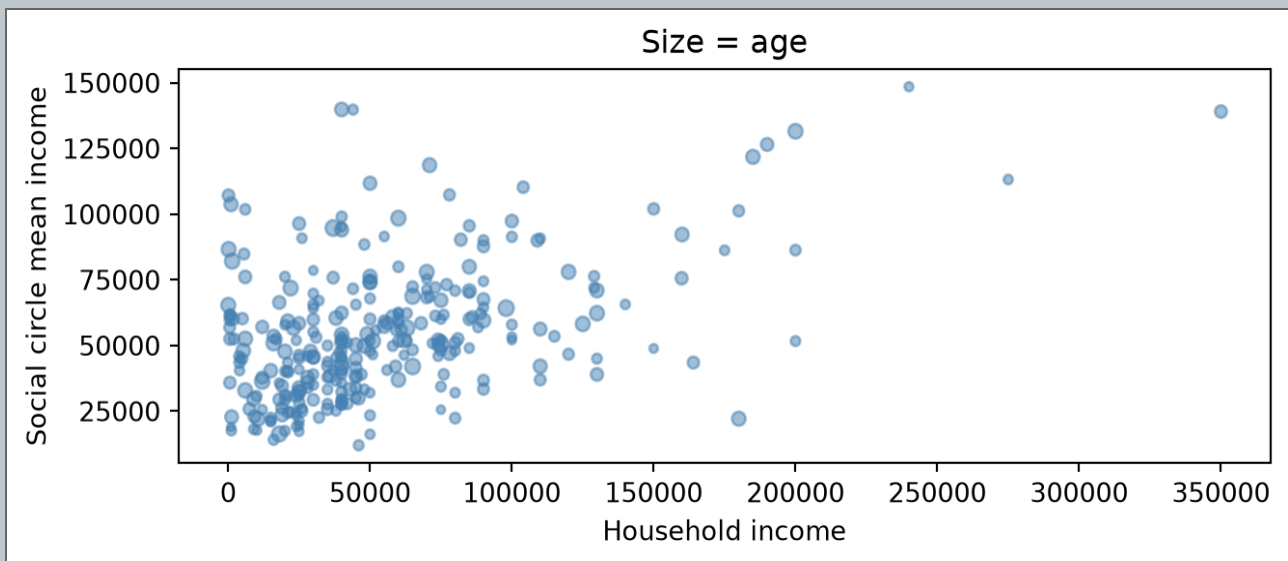
Visual variables (Bertin's semiology)

Visual variable	Best for	Example
Position (x, y)	primary variables	the main relationship
Size	third numeric variable	population, importance
Colour (hue)	categories	groups, types
Colour (saturation)	continuous values	intensity, amount
Shape	categories	different types
Transparency / jitter	overlapping points	density

Every visual element should encode information.

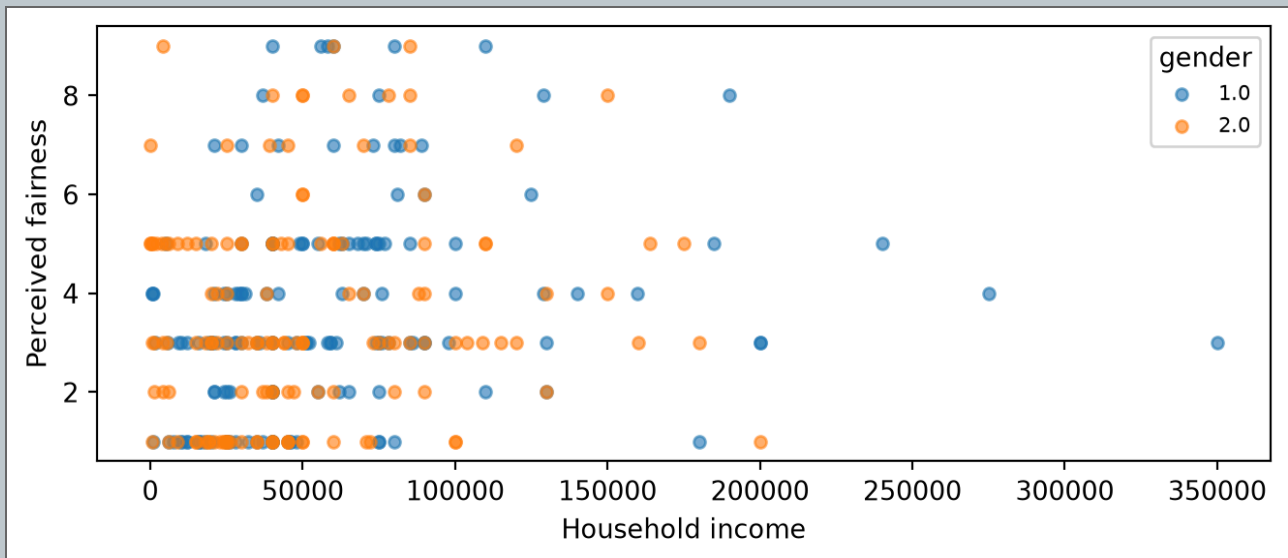
A third variable: size and transparency

```
1 sub = dawtry[["Household_Income", "Social_Circle_Mean_Income", "age"]].dropna()
2
3 fig, ax = plt.subplots(figsize=(7, 3))
4 ax.scatter(sub["Household_Income"], sub["Social_Circle_Mean_Income"],
5           s=sub["age"] / 2, alpha=0.5, color="steelblue")
6 ax.set_xlabel("Household income")
7 ax.set_ylabel("Social circle mean income")
8 ax.set_title("Size = age")
9 plt.tight_layout()
10 plt.show()
```



A third variable: colour by category

```
1 sub = dawtry[["Household_Income", "fairness", "gender"]].dropna()
2
3 fig, ax = plt.subplots(figsize=(7, 3))
4 for name, grp in sub.groupby("gender"):
5     ax.scatter(grp["Household_Income"], grp["fairness"], label=str(name), alpha=0.6, s=
6 ax.set_xlabel("Household income")
7 ax.set_ylabel("Perceived fairness")
8 ax.legend(title="gender", fontsize=8)
9 plt.tight_layout()
10 plt.show()
```



Part V: Thinking about causes

Extra, if time

Types of patterns

1. Differences between groups
2. Social change
3. Cross-national comparisons

Figure 2. Leave vote by objective characteristics



http://natcen.ac.uk/media/1319222/natcen_brexplanations-report-final-web2.pdf

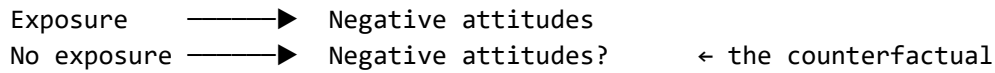
Causal inference

Exposure to hateful or toxic content on a platform —————> Negative attitudes toward minority groups
hypothetical causal relationship

Language: the first is the *independent variable* (IV), also called causal, explanatory, or predictor. The second is the *dependent variable* (DV), also outcome, symptom, or response.

Counterfactuals

To claim the first causes the second, you need to know what would have happened **without** it.



The diagram shows two horizontal rows of text. The top row is 'Exposure' followed by a right-pointing arrow and 'Negative attitudes'. The bottom row is 'No exposure' followed by a right-pointing arrow and 'Negative attitudes?'. To the right of the bottom row, there is a left-pointing arrow followed by the text '← the counterfactual'. A vertical blue line is on the left side of the diagram.

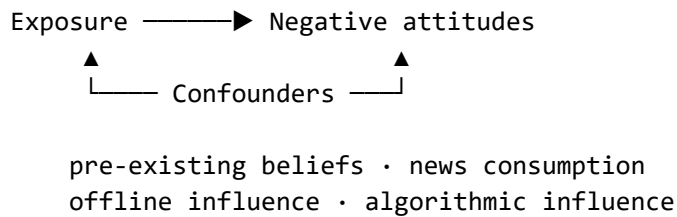
```
graph LR; A[Exposure] --> B[Negative attitudes]; C[No exposure] --> D[Negative attitudes?]; D --> E[← the counterfactual]
```

Exposure —————▶ Negative attitudes
No exposure —————▶ Negative attitudes? ← the counterfactual

You cannot observe both for the same person. That is the fundamental problem of causal inference.

Confounding variables

A working counterfactual is not always enough.



Hill, “Contextual Analytics”, §5.4–5.5 — the mechanisms that remain unrepresented must not be silently absorbed into the interpretation.

Part VI: Live Coding

Live coding

1. Scatter plot of two variables from the survey data
2. Pearson and Spearman, and why they differ here
3. A third variable by size, then by colour
4. A null result, and how to report it
5. Save the figure

Before next week

1. Complete `w08_worksheet.ipynb` in the workshop
2. Read Hill, “Contextual Analytics”, §5.4–5.5

Note

The quiz in next week’s lecture covers **this week’s** material.



Speaker notes